DataCite Corpus - Data Exploration & Descriptive Statistics

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DataCite Corpus - Exploratory Analysis

```
## FUNCTION: exp_summary - argument dat (dataframe)
## Argument dat takes a dataframe. Column names are assumed to
## match the columns provided in the DataCite Corpus as of
## 11 October 2026. Documentation and file downloads available
## at https://zenodo.org/records/16901115

dc_summary <- function(dat) {
  ## Returns descriptive statistics of the data file - number
  ## of rows, number & proportion of rows with publication DOIs,
  ## number & proportion of rows with a 'created' date,
  ## number & proportion of rows with publication dates,
  ## number & proportion of rows with a funder listed.

  summarylist <- dat |> summarise(
    n = nrow(dat),
    n_doi = str_detect(dat$publication, pattern = "doi") |> sum(na.rm = TRUE),
    n_created_date = sum(1 - is.na(dat$created), na.rm = TRUE),
    n_pub_date = sum(1 - is.na(dat$publishedDate), na.rm = TRUE),
    n_funders = sum(1 - is.na(dat$funders), na.rm = TRUE),
    n_nih = str_detect(dat$funders, pattern = "(NIH)|(National Institutes of Health)") |>
      sum(na.rm = TRUE)
  )

  return(summarylist)
}
```

```
## FUNCTION: do_dc_summary - wrapper for dc_summary for
## calling with %dopar%.
```

```
## FUNCTION: get_nih, reads in dataframe, perform basic preprocessing,
## return only data for NIH-funded datasets (based on fundersROR).

get_nih <- function(filename) {
  dat <- read_csv(file = filename, na = c("", "NA", "none"))

  ## Prints column names and types for debugging.
  c("Columns:", colnames(dat)) |> paste(collapse = ", ") |> print()
  c("Column types", sapply(dat, typeof) %>% unlist()) |>
    paste(collapse = ", ") |> print()

  ## Sets date and datetime columns to POSITX type.
```

```
  dat <- dat |> mutate(
    across(c(created, updated), as_datetime)
    )
  dat <- dat |> mutate(
    across(publishedDate, as.Date)
    )

  # Makes all character columns (excluding 'id', 'dataset', and 'publication') uppercase.
  dat <- dat |> mutate(
    across(c(repository:title, source:fundersROR), toupper)
    )

  # Append summary data and data from NIH-funded datasets.
  return(dat[which(
    str_detect(dat$funders, pattern = "(NIH)|(National Institutes of Health)")
    ),]
    )
}
```

```
# Function computing descriptive stats
# TODO: parallelize

## FUNCTION: read_datacite - arguments dirpath (string)
## Argument dirpath takes a file path to the directory containing
## the DataCite Corpus files to be read.
read_datacite <- function(filenames) {
  ## For each file in the directory, output column names and types,
  ## convert numeric columns to date/datetime, and convert string columns
  ## to uppercase. Call exp_summary on each dataset, add output values to
  ## summary_data, and add NIH-funded datasets to df.

  cl <- makeCluster(4)
  registerDoParallel(cl)

  nih_df <- foreach(fn = files,
                    .combine = 'rbind',
                    .export = 'get_nih',
                    .packages = 'tidyverse') %dopar% {
    get_nih(fn)
  }

  stopCluster(cl)

  # Return data from NIH-funded datasets only.
  return(nih_df)
}
```

```
files <- list.files(path = paste0(dirpath, "/data"),
                    full.names = TRUE, recursive = TRUE)

# Pull NIH data from DataCite Corpus
df <- read_datacite(fns)
head(df, 20)
```

```
## # A tibble: 20 x 16
##    id       created             updated             repository publisher journal
##    <chr>    <dttm>              <dttm>              <chr>      <chr>     <chr>
##  1 009813e~ 2023-06-07 01:53:40 2023-06-07 01:53:40 UNIVERSIT~ CNRI ACCT D-LIB ~
##  2 00c4816~ 2023-06-07 01:53:40 2023-06-07 01:53:40 UNIVERSIT~ SPRINGER~ TROPIC~
##  3 00f019e~ 2025-07-24 17:16:04 2025-07-24 17:16:04 DUKE RESE~ AMERICAN~ MICROB~
##  4 00f8812~ 2025-07-24 21:21:02 2025-07-24 21:21:02 UNIVERSIT~ SOCIETY ~ JOURNA~
##  5 010a142~ 2023-06-07 11:40:02 2023-06-07 11:40:02 UNIVERSIT~ COLD SPR~ <NA>
##  6 0187d97~ 2025-01-31 00:17:18 2025-01-31 00:17:18 HARVARD D~ AMERICAN~ JOURNA~
##  7 020b7a3~ 2023-06-07 10:38:06 2023-06-07 10:38:06 DRYAD      WILEY     JOURNA~
##  8 02187b0~ 2025-01-31 00:55:08 2025-01-31 00:55:08 HARVARD D~ COLD SPR~ <NA>
##  9 02949d9~ 2023-06-08 18:59:45 2023-06-08 18:59:45 DRYAD      PROCEEDI~ PROCEE~
## 10 02c7b62~ 2025-07-25 16:32:03 2025-07-25 16:32:03 UNIVERSIT~ ELSEVIER~ DATA I~
## 11 034d42f~ 2025-01-31 00:25:49 2025-01-31 00:25:49 HARVARD D~ AMERICAN~ JOURNA~
## 12 03937ee~ 2025-07-25 02:21:28 2025-08-13 19:11:43 USC ISI S~ PROCEEDI~ PROCEE~
## 13 039e011~ 2023-06-06 16:39:24 2023-06-06 16:39:24 DRYAD      OXFORD U~ TOXICO~
## 14 0449dfa~ 2023-06-08 20:01:42 2023-06-08 20:01:42 DRYAD      ELIFE SC~ ELIFE
## 15 048d16a~ 2025-07-23 22:50:35 2025-08-13 16:13:24 GUDMAP (W~ THE COMP~ BIOLOG~
## 16 054c1da~ 2025-07-24 08:40:48 2025-08-13 16:42:46 IMMPORT    MDPI AG   VIRUSES
## 17 05bdf54~ 2025-07-24 15:32:24 2025-07-24 15:32:24 APOLLO - ~ THE ENDO~ THE JO~
## 18 05df4b9~ 2025-07-24 21:21:02 2025-07-24 21:21:02 UNIVERSIT~ SOCIETY ~ JOURNA~
## 19 06041aa~ 2025-01-30 23:41:53 2025-01-30 23:41:53 HARVARD D~ AMERICAN~ JOURNA~
## 20 0662231~ 2025-07-25 16:37:40 2025-08-13 20:44:32 FACEBASE ~ THE COMP~ DEVELO~
## # i 10 more variables: title <chr>, dataset <chr>, publication <chr>,
## #   publishedDate <date>, source <chr>, subjects <chr>, affiliations <chr>,
## #   affiliationsROR <chr>, funders <chr>, fundersROR <chr>
```

```
dat <- read_csv("/Users/ashvin/Desktop/biomed-retractions/datacite/data/2025-08-15-data-citation-corpus
```

```
# Data frame to hold data from NIH-funded datasets.
  nih_df <- data.frame(
    "id" = c(),
    "created" = c(),
    "updated" = c(),
    "repository" = c(),
    "publisher" = c(),
    "journal" = c(),
    "title" = c(),
    "dataset" = c(),
    "publication" = c(),
    "publishedDate" = c(),
    "source" = c(),
    "subjects" = c(),
    "affiliations" = c(),
    "affiliationsROR" = c(),
    "funders" = c(),
    "fundersROR" = c()
  )

  # Summary data across all datasets.
  data_summary <- data.frame(
```

```
    "n" = c(),
    "n_doi" = c(),
    "n_created_date" = c(),
    "n_pub_date" = c(),
    "n_funders" = c(),
    "n_nih" = c(),
    "n_nih_ror" = c()
  )


# Compute summary descriptive statistics across summary_data.
  data_summary |> summarise(
    Size = sum(n),
    Num.DOI = sum(n_doi),
    Percent.DOI = (sum(n_doi) / sum(n)) * 100,
    Num.created = sum(n_created_date),
    Percent.created = (sum(n_created_date) / sum(n)) * 100,
    No.funders = (sum(n_funders)),
    Percent.funders = (sum(n_funders) / sum(n)) * 100,
    No.NIH.Funded = sum(n_nih)
    No.
  )

dat_samp <- dat[sample(1:nrow(dat), 500),]
```